DAY - 16 SEAT NUMBER

2023	VIII	07	1100	V – 198	(E)

COMPUTER SCIENCE
PAPER - II (D-9)

Time : 3 Hours | 4 Pages | Max. Marks : 50

Instructions : (1) *All question are compulsory.*
(2) *Figures to the right indicate full marks.*
(3) *Draw neat diagram wherever necessary.*
(4) *Use of any type of calculator is not allowed.*
(5) *Draw neat diagrams, wherever necessary.*

1. (A) Select the correct alternative and rewrite the following :

(a) The 8085 Microprocessor can address ________ bytes of memory. 1

(i) 4 kilo
(ii) 1 kilo
(iii) 32 kilo
(iv) 64 kilo

(b) The ________ instruction rotates contents of accumulator left through carry by one bit. 1

(i) RAL
(ii) RLC
(iii) RRC
(iv) RAR

(c) The 8051 has contain ________ special function registers. 1

(i) 8
(ii) 16
(iii) 24
(iv) 22

(d) Ethernet network transmit data in small units called ________. 1

(i) byte

(ii) nibble

(iii) frame

(iv) word

(B) Answer **any two** of the following :

(a) State any two advantages of computer networks. Distinguish between LAN and WAN. (Any four points) 3

(b) Define following terms in transmission media : 3

(i) Bandwidth

(ii) Attenuation

(iii) Immunity from electromagnetic interference.

(c) Write the use of following terms in functional block diagram of 8085 Microprocessor : 3

(i) Serial I/O controller

(ii) Timing and Control unit

(iii) ALU

2. (A) Answer **any two** of the following :

(a) Draw flag register diagram of 8085 Microprocessor. Interpret its meaning by taking your own example. 3

(b) Draw neat labelled memory map diagram for 8051 microcontroller. 3

(c) State any three advance features of X-86 Microprocessor family. 3

(B) Answer **any one** of the following :

(a) Define STAR topology with diagram. Write its two advantages and disadvantages. 4

(b) State any four addressing modes with one example each in 8085 Microprocessor. 4

3. (A) Answer **any two** of the following :

(a) Write any six features of 8085 Microprocessor. 3

(b) Define Software Interrupt. Write all vector addresses (call locations) of software interrupts. 3

(c) Write a note on complete microprocessor system (i.e. microcontroller). 3

(B) Answer **any one** of the following :

(a) After execution of following instruction in 8085 Microprocessor mention which flag(s) gets affected : 4

(i) INR M

(ii) DCX rp

(iii) DAD rp

(iv) CMA

(b) Explain the advantages of Pentium Processor with respect to the following features : 4

(i) Dual pipelining

(ii) On chip caches

(iii) Branch prediction

(iv) 64 bit data bus

4. (A) Answer **any two** of the following :

(a) Write addressing mode and length in bytes for following 8085 instructions : 3

(i) ADI 29H

(ii) XCHG

(iii) LDAX B

(b) Write single instruction in 8085 Microprocessor which clears (00H) the content of Accumulator after execution. 3

(c) What is Router ? Explain its two types. 3

(B) Answer **any one** of the following :

(a) What is Co-axial Cable ? Draw labelled diagram for it. State its two advantages and disadvantages. 4

(b) State functions of following pins in 8085 microprocessor :

(i) $AD_0 - AD_7$

(ii) $A_8 - A_{15}$

(iii) S_0 and S_1

(iv) V_{cc} and V_{ss}

5. Answer **any two** of the following :

(a) Write an assembly language program to find absolute difference of two hex numbers stored in memory locations 5002H and 5003H. Store the result at 5004H. 5

(b) A block of data is stored in memory locations from 5004H. The length of block is stored at memory location 5003H. Write an assembly language program that searches for the first occurrence of data byte 2AH in the given block. Store the address of this occurrence in HL register pair. If number is not found then HL register pair must contain FFFFH. 5

(c) The 8 bit hex number is stored in register B if both nibbles are having same value then store 00H in accumulator else FFH in accumulator. 5

OR

5. Answer **any two** of the following :

(a) Write an assembly language program using rotate instruction to divide the number in BC register pair by 2 and store result in HL register pairs. 5

(b) Write an assembly language program that adds 4 byte integer stored in consecutive memory locations starting from C200H beginning with lower order byte to another 4 byte integer stored in consecutive memory locations starting from C300H beginning with lower order byte. Store the result in consecutive memory location starting from C200H. 5

(c) Write an assembly language program to convert a two digit BCD [Binary Coded Decimal] number stored at memory location B204H into its binary equivalent and store the binary value in memory location B205H. 5
